

SCAN CONVERSION

The screen of a computer is divided into rows and columns. The intersection area of these rows and columns is known as a pixel. Scan conversion is the process of creating a pattern of points identical to the objects to be displayed. Each pixel on the display surface has a finite size depending on the screen resolution and hence a pixel cannot represent a single point. However, we consider each pixel as unit square area identified by the coordinate with respect to lower (or upper) left corner.

SCAN-CONVERTING A POINT

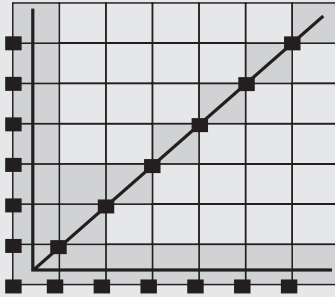
A mathematical point (x, y) where x and y are real numbers within an image area, needs to be scan converted to a pixel at location (x', y') . This may be done by making x' to be the integer part of x and y' the integer part of y . In other words, $x' = \text{Floor}(x)$ and $y' = \text{floor}(y)$, where function Floor returns the largest integer that is less than or equal to the argument. All points that satisfy $x' \leq x < x' + 1$ and $y' \leq y < y' + 1$ are mapped to a pixel (x', y') . For example, point $P_1 (1.7, 0.8)$ is represented by pixel $(1, 0)$. Points $P_2 (2.2, 1.3)$ and $P_3 (2.8, 1.9)$ are both represented by pixel $(2, 1)$.

In another approach we can convert (x, y) by making $x' = \text{Floor}(x + .5)$ and $y' = \text{Floor}(y + .5)$. All points that satisfy $x' - 0.5 \leq x < x' + 0.5$ and $y' - 0.5 \leq y < y' + 0.5$ are mapped to pixel (x', y') . This means that points P_1 and P_2 are now both represented by pixel $(2, 1)$, whereas point P_3 is represented by pixel $(3, 2)$.

SCAN CONVERTING A LINE

In order to draw a line on the screen, first we have to determine which pixels are to be switched ON. The process of determining which combination of pixels provides the best approximation to the desired line is known as **Rasterization**. The choice of pixels is determined by the orientation of the line which is to be drawn. There is little difficulty in selecting the pixels for horizontal, vertical and diagonal lines. For any other orientation of lines, selection of pixels is a difficult process.

For horizontal, vertical and 45° lines, the choice of raster elements is obvious. This lines exhibit constant brightness along the length:



For any other orientation the choice is more difficult:

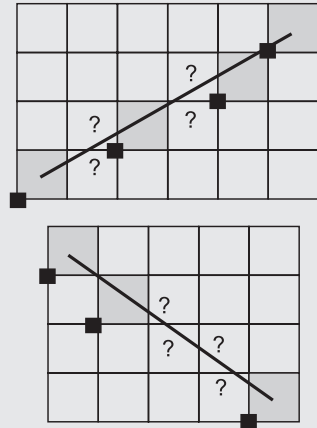


Fig. 2.1. Rasterization process for drawling lines

General requirement for drawing a line

- Lines must appear to be straight
- Lines should start and end accurately
- Lines should have constant brightness along their length.
- Lines should be drawn rapidly.

On the raster scan systems, lines are plotted with pixels and step sizes in the horizontal and vertical directions are constrained by pixel separations. We must sample a line at discrete positions and determine the nearest pixel to the line at each sample position. This is decided by the equation of a straight line.

Equation for a straight line is,

$y = mx + c$ where m is the slope of the line and c is the y-intercept

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$$

DDA Algorithm

Let the line is to be drawn between the end points $(x_{\text{start}}, y_{\text{start}})$ and $(x_{\text{end}}, y_{\text{end}})$. Let the equation of the line as $y = mx + c$ where m is slope of the line and c is the intercept. The slope can be expressed as

$$m = \frac{y_{\text{end}} - y_{\text{start}}}{x_{\text{end}} - x_{\text{start}}}$$

In fact any two consecutive points (x_i, y_i) and (x_{i+1}, y_{i+1}) lying on this line segment should satisfy the equation $m = \frac{y_i + 1 - y_i}{x_i + 1 - x_i}$ if we say $y_{i+1} - y_i = \Delta y$ and $x_{i+1} - x_i = \Delta x$ then $\frac{\Delta y}{\Delta x} = m$

$$\text{So} \quad \Delta y = m \Delta x$$

Thus, for a given x interval Δx along the line, we can compute the corresponding y interval y . Now if $\Delta x = 1$ i.e. $x_{i+1} - x_i = 1$ or $x_{i+1} = x_i + 1$ then $\Delta y = m$ which implies $y_{i+1} = y_i + m$.

Thus, a unit change in x changes y by m which is a constant for a given line. We know that if $x_{i+1} = x_i + 1$, then $y_{i+1} = y_i + m$, the values of x and y are defined in terms of their previous values. Initializing (x_i, y_i) as $(x_{\text{start}}, y_{\text{start}})$ the line can be generated by incrementing the previous x values by one unit and solving the corresponding y value at each step till x_{end} is reached. At each step we make incremental based on the previous step. This is what is defined as incremental algorithm and often referred to as the **Digital Differential Analyzer (DDA)** algorithm.

A line segment is to be drawn from (x_1, y_1) to (x_2, y_2)

Generating a line segment is equivalent solving a simple differential equation:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$$

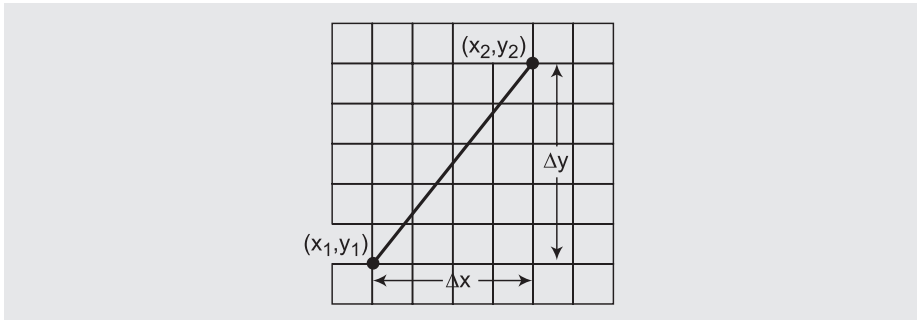


Fig. 2.2. Digital Differential Analyzer (DDA) Algorithm

Starting from (x_1, y_1) , for any change in x of size Δx , the corresponding change in y is:

$$\Delta y = m \Delta x$$

While incrementing the values of y by constant m , in this DDA method we have to keep in mind that m being the slope of the line, it can be any real number negative or positive – so that the calculated y values must be round off to the nearest integer for the sake of plotting on the screen.

But the question is why not increment y values by 1 instead of x and accordingly calculate x values from the line's equation. The determining factor here is absolute value of the slope of the line $|m|$. If $|m| \leq 1$ which implies $|y| \leq |x|$ we sample the line

at unit x intervals. But if $|m| > 1$ means $|y| > |x|$ a unit step in x creates a step in y that is greater than 1, which is not desirable. In that case we reverse the roles of x and y by sampling at unit y intervals as

$$\Delta y = y_{i+1} - y_i = 1 \text{ and } dx \frac{\Delta y}{m}, x_{i+1} = x_i + \frac{1}{m}$$

Both the equations are based on the assumption that $x_{\text{start}} < x_{\text{end}}$ and $y_{\text{start}} < y_{\text{end}}$ i.e. slope is positive. But if it is not so then we have to apply negative increment

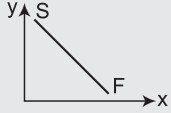
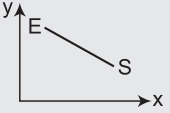
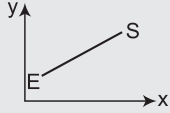
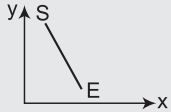
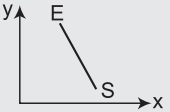
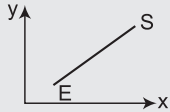
	$X_{\text{start}} < X_{\text{end}}$ $Y_{\text{start}} > Y_{\text{end}}$ $\Rightarrow m$ is negative	$X_{\text{start}} > X_{\text{end}}$ $Y_{\text{start}} < Y_{\text{end}}$ $\Rightarrow m$ is negative	$X_{\text{start}} > X_{\text{end}}$ $Y_{\text{start}} > Y_{\text{end}}$ $\Rightarrow m$ is positive
$ m \leq 1$	 $X_{i+1} = X_i + 1$ $Y_{i+1} = Y_i + m$	 $X_{i+1} = X_i - 1$ $Y_{i+1} = Y_i - m$	 $X_{i+1} = X_i - 1$ $Y_{i+1} = Y_i - m$
$ m \geq 1$	 $Y_{i+1} = Y_i - 1$ $X_{i+1} = X_i - \frac{1}{m}$	 $Y_{i+1} = Y_i + 1$ $X_{i+1} = X_i - \frac{1}{m}$	 $Y_{i+1} = Y_i - 1$ $X_{i+1} = X_i - \frac{1}{m}$

Fig. 2.3. Orientation of the line according to the slope.